WIUT HACKATHON 2026 · COMPUTER VISION FINAL

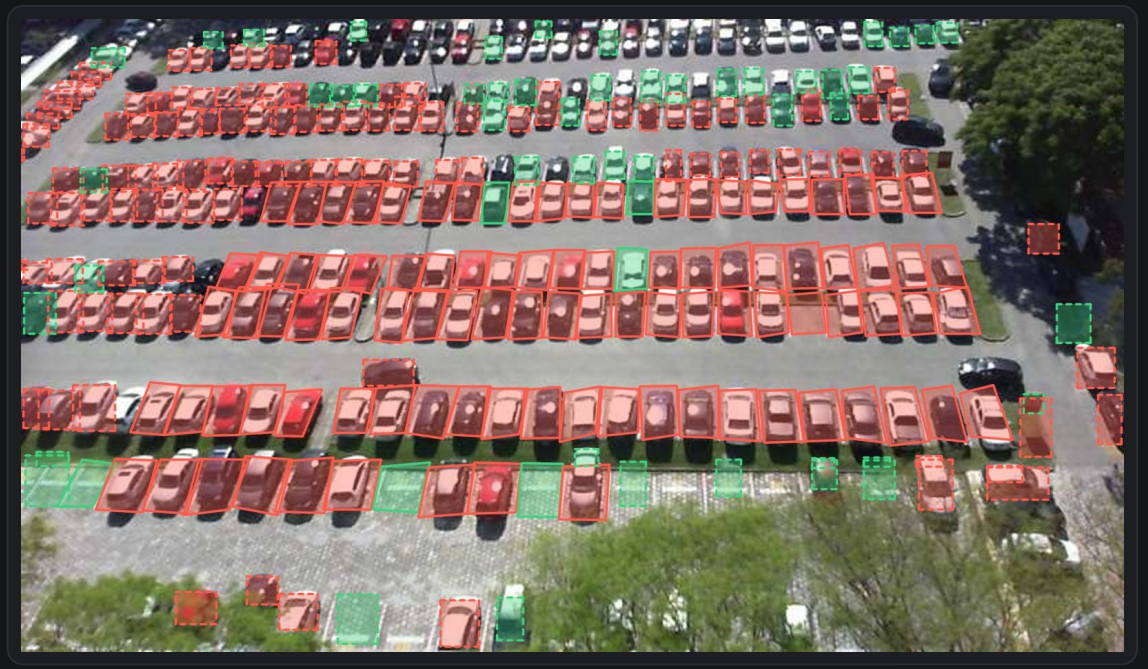
OQIM

A parking tracker from car park cameras

MODE: STARTUP

Team PariVision
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Live https://oqim-audit.vercel.app/tracker
Code https://github.com/baytoooo/oqim



The PUCPR car park camera at 09:04 on 17 October 2012 (PKLot dataset). Green: free, red: taken, as our detector read the frame; dashed: spaces the camera found outside the hand labels.

PUCPR | 09:04 | 55 free of 260

Live today: /tracker /parking /city /banner

PROBLEM AND USER

Drivers circle because they cannot see free spaces

17 h

a year a US driver spends searching for parking (INRIX, 2017)

~5%

of off-street car parks publish live free spaces (Parkopedia)

13,208

paid spaces on 86 streets planned by Poytaxt Parking, Tashkent

SFpark in San Francisco cut cruising for parking by 50% on its pilot blocks. Sources: INRIX, Parkopedia, Spot.uz, ACCESS Magazine

How a driver finds a space today

- Picks a car park by distance or habit, and learns it is full at the gate.
- Drives on to the next one. While searching, the car is part of the traffic.
- A navigator sends every driver to the same nearest car park.

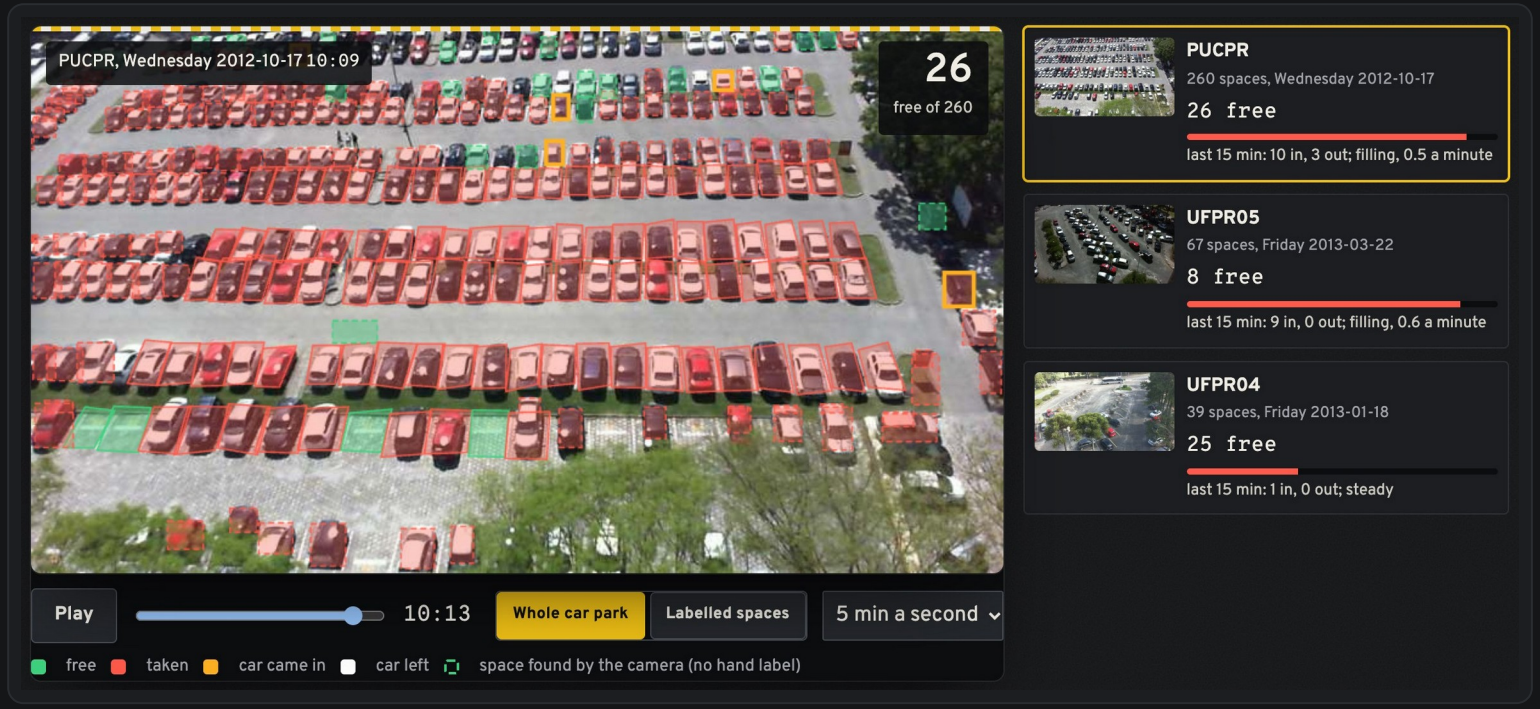
The user, and who pays

User: drivers going to malls, offices, hospitals and stations.

Who pays:

- car park operators: fewer drivers turned away
- the city: less searching traffic
- outdoor advertising: traffic past each billboard

Free spaces now, and when you arrive



Ask now: where should I park?

Drive to PUCPR

About 23 spaces free when you get there at 10:19, a 100% chance of a space.

Car park	Free now	Expected on arrival	Chance of a space	Minutes to the door
PUCPR (go)	26	23	100%	8.0
UFPR05	8	7	84%	13.3
UFPR04	25	24	100%	14.0

▶ Your trip (example times, change them)

Many drivers asking at the same time

Drivers 40

Every navigator: the nearest car park

17 of 40

find it full. PUCPR 40. 11.4 min to the door on average.

OQIM: one by one, counting the drivers already sent

4 of 40

find it full. PUCPR 27, UFPR05 3, UFPR04 10. 10.6 min to the door on average.

1 **Play the morning**

2 **Whole car park**

3 **One driver asks**

4 **40 drivers ask**

https://oqim-audit.vercel.app/tracker 3 PKLot cameras, one clock, 06:55 to 10:35, a frame every 5 min

Labelled spaces: PUCPR 100 (2012-10-17), UFPR05 40 (2013-03-22), UFPR04 28 (2013-01-18)

At 10:13, 40 drivers: the nearest car park leaves 17 of 40 at a full gate, OQIM 4 of 40.

98.8% of space readings right

98.8%

of 6,844 space readings right, 112 frames of 3 cameras

1 / 4

spaces the free count is off by, median / 90th percentile, per frame

+198 spaces

found outside the 168 hand-labelled ones, from where cars stay 20 min

139 of 142

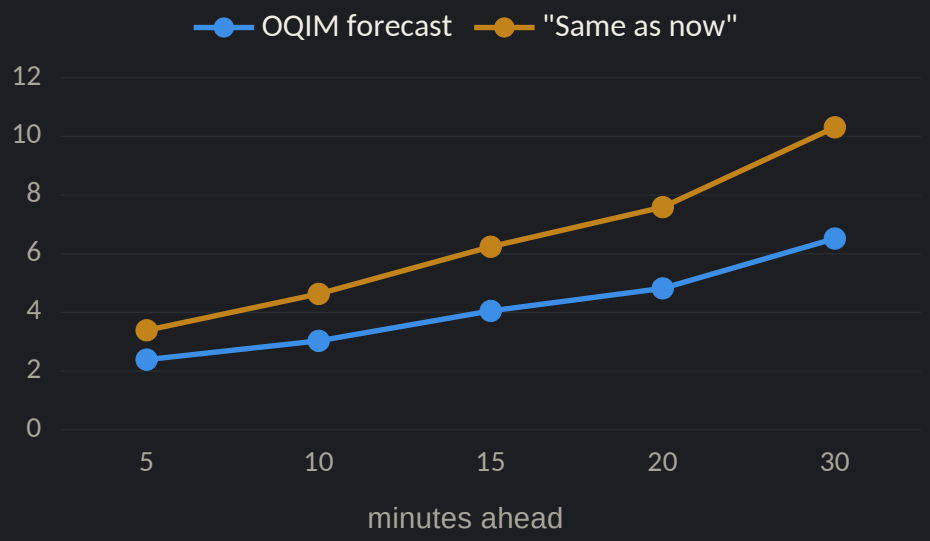
labelled spaces that cars used, found the same way: the check for the method

How it was checked: every frame against the dataset's hand labels. Same rule as /parking, picked on its 96 calibration images. Cars seen on the labelled spaces: 146 in, 11 out.

Be clear: recorded mornings of a public dataset, not a live feed yet.

Forecast error by minutes ahead

mean spaces off, three car parks



10 min ahead: 3.0 spaces off, against 4.6 for "same as now", over 101 forecasts.

Trend: half the last 20 minutes, half the car park's typical weekday, from the labels of its other days.

From a camera frame to advice

For each camera, the same pipeline on every page

Frames

One frame every 5 min from an ordinary fixed camera, or a photo, or a video.

Detect

YOLO26m on the tracker frames; YOLO26n (10 MB) in the browser. Whole frame plus 3 x 3 tiles at 960 px.

Space state

Taken when one box covers 40% of the space. It changes only when two frames in a row agree.

Forecast

Free spaces, cars in and out. Free spaces 10 min ahead from the last 20 minutes and the usual weekday.

Advise

Drivers placed one by one on the car park and route of least door-to-door time, counting those already sent.

10 MB

detector, YOLO26n in ONNX Runtime Web: WebGPU or WebAssembly

0

GPU servers. The detector runs in a browser tab.

No plates

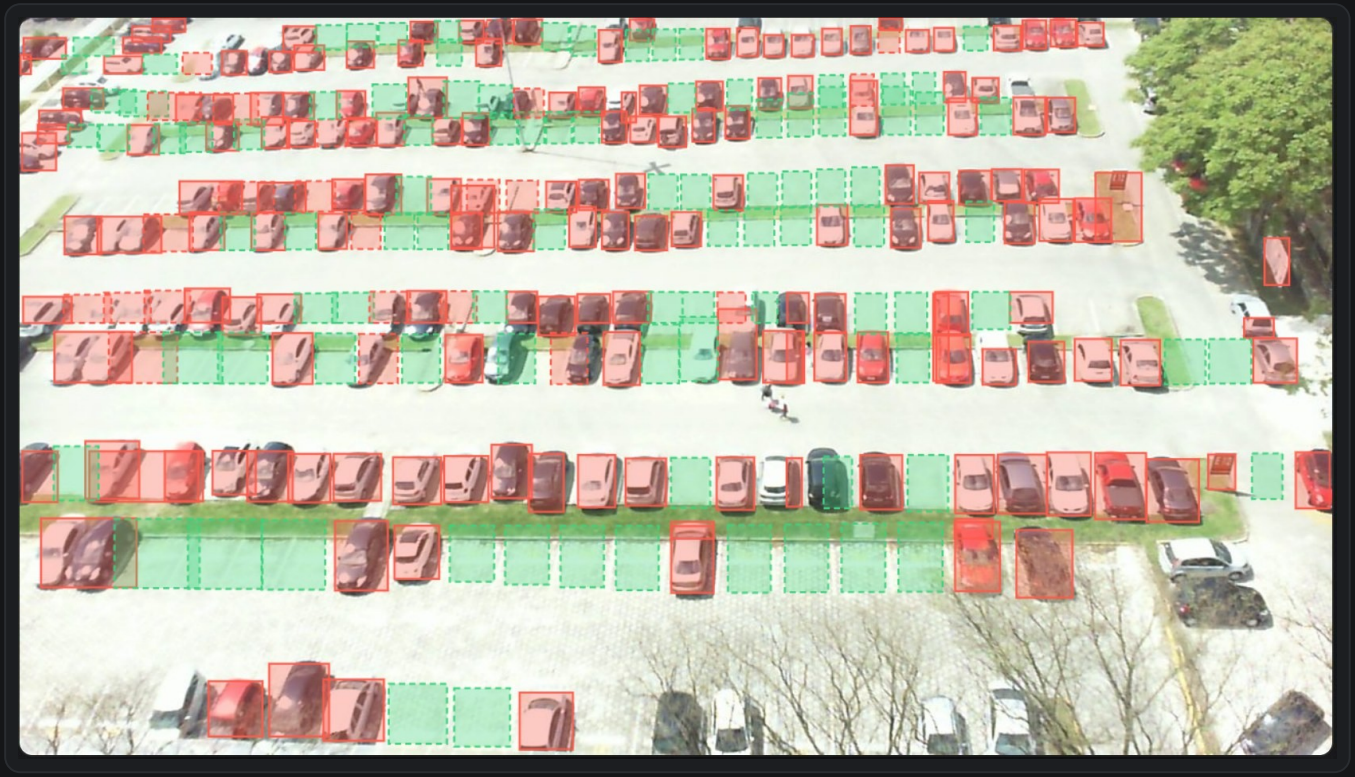
and no faces are read. Only vehicle boxes.

4 pages

one pipeline: tracker, own photo, district, billboard counter

LIVE DEMO 2 OF 4 · /parking

Your own photo, no hand marking



Red: every car found is a taken space. **Dashed green:** a gap in a row of cars that fits a car is a suggested free space.

A PKLot frame loaded as your own photo, no hand labels used. Far rows are also read on 5 x 5 tiles, each enlarged.

88.4%
of taken spaces found from the photo alone

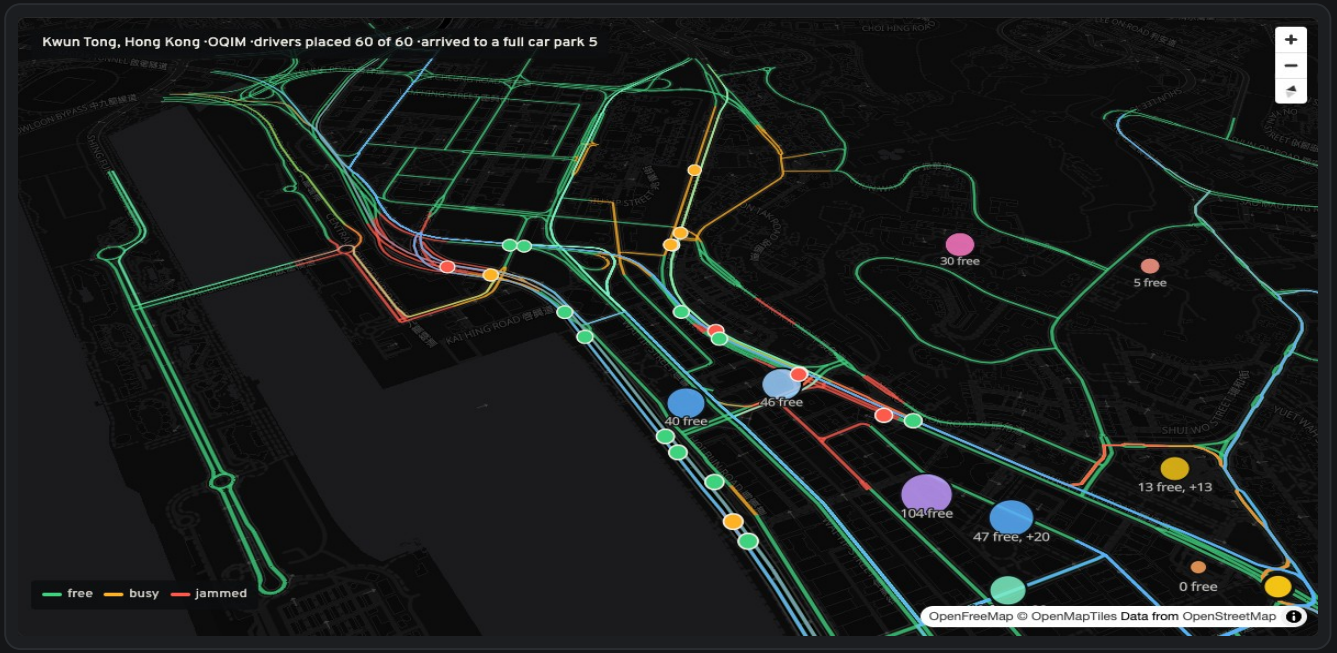
83.5%
of suggested free spaces really are free

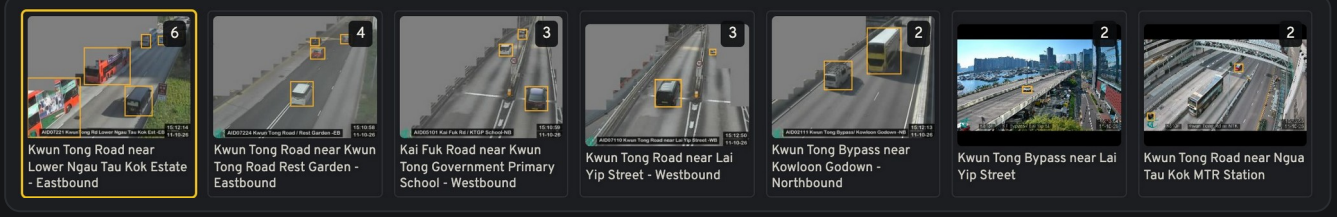
Checked on 336 PKLot images: the same images its settings were picked on. Not held out.

97.4%
of 13,447 spaces right on 240 held-out images when the spaces are marked; the rule was picked on 96 other images

Limit: it suggests 12.2% of all free spaces. A free space with no car on either side, such as an empty row, cannot be seen in one photo: draw it with 4 clicks.

A district: 22 live cameras and car parks





Transport Department cameras (DATA.GOV.HK), 1,116 snapshots counted by YOLO26m; boxes in orange. Live vacancy of 28 car parks.

60 → 5

of 60 drivers arrive to a full car park: nearest car park against OQIM

−18%

door to door, 15.5 to 12.7 min

−41%

vehicle-km with searching, 305 to 181

How: drivers are placed one by one. Each gets the car park and route of least door-to-door time, with the roads and spaces already taken by the drivers before.

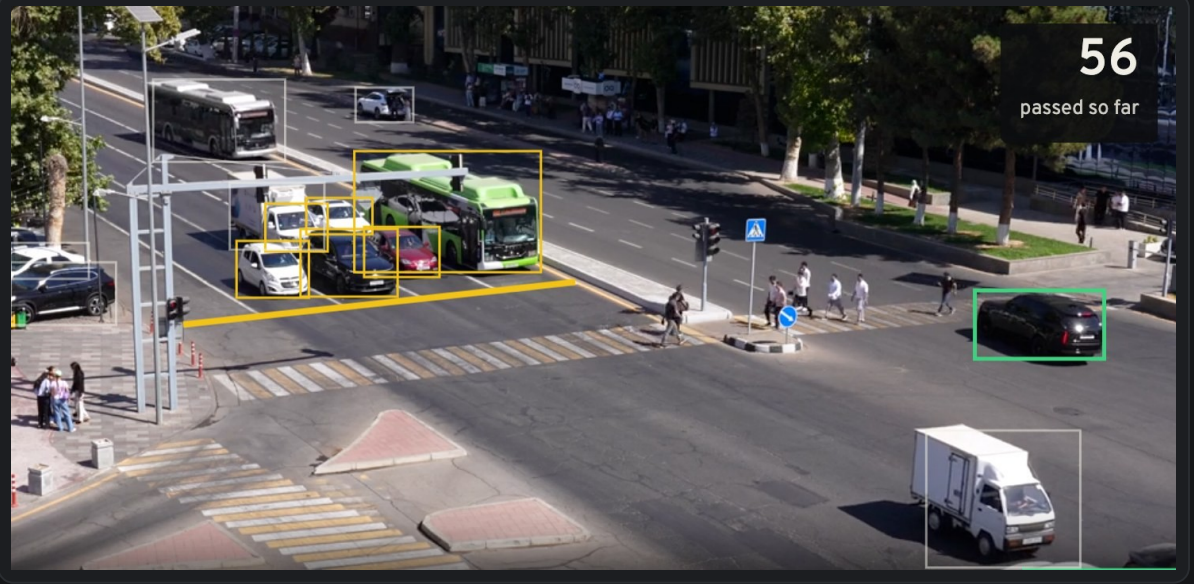
Measured: camera counts and car park vacancy.

Simulated: the drivers.

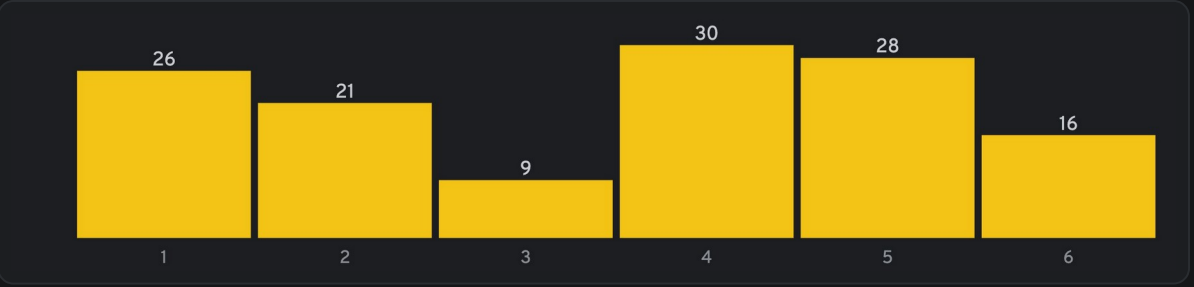
Snapshot: 13:45 Hong Kong time, 11 October. The live page shows today's numbers.

NEW · LIVE DEMO 4 OF 4 · /banner

How many vehicles pass your billboard



Vehicles per minute past the line, sample C3896 (5 min 40 s)



Draw one line in front of the billboard

- each tracked vehicle is counted once, when it crosses the line
- by type (car, bus, truck, motorcycle) and direction
- per minute and per hour; a log per billboard; CSV

185 vs 189

vehicles on 2 sample clips: YOLO26s in the browser against our offline YOLO26m pipeline, 97.9%

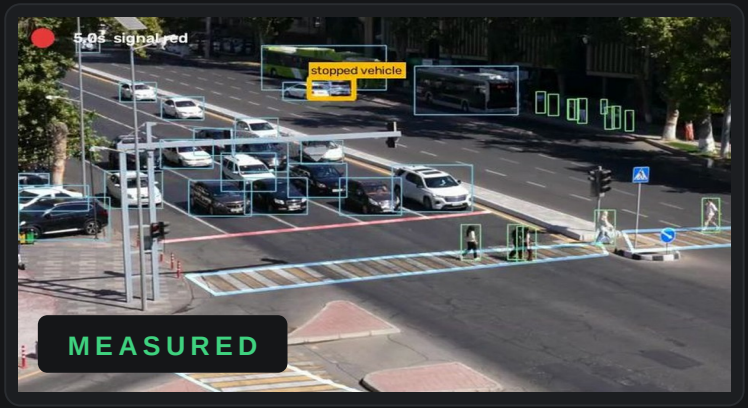
C3896 (midday, 5:40): 130 against 130; C3905 (dusk, 2:08): 55 against 59

Two models agreeing, not a hand count. A hand count is still to do.

Who pays: outdoor advertising companies, to show advertisers the traffic past each billboard, hour by hour, from a fixed camera or a phone video.

NEXT STAGE · /audit, /network AND /roads

Next: the same cameras time the lights

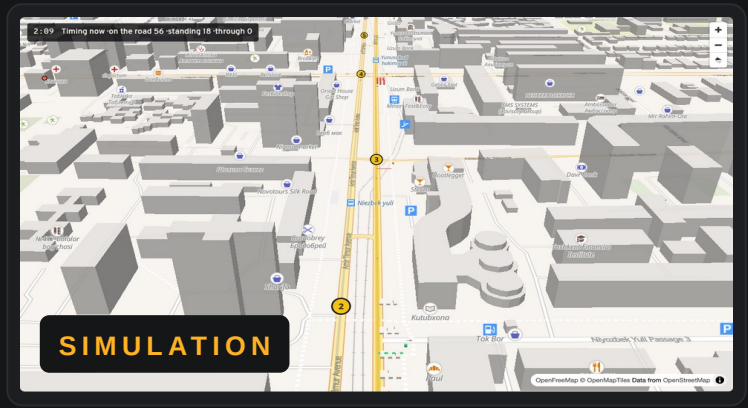


Signal audit · /audit

The lights read from pixels, every vehicle over the stop line, signal plans checked on a replay. 4 clips of one Tashkent junction (organisers' footage).

0.29 to 0.55 s
replay error on 3 of 4 clips; C3905 too short, fails

4 of 4
clips where the textbook Webster plan is worse than now

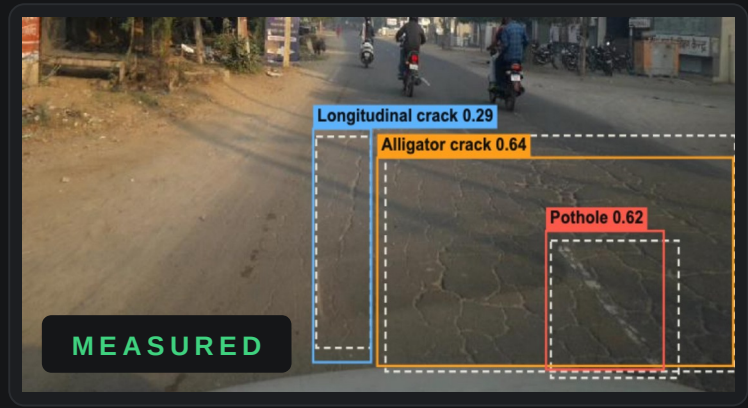


Green wave · /network

Amir Temur Avenue, Tashkent: 1.46 km, 5 signals, 224 vehicles in 10 min. Only the offsets change.

141 s → 24 s
mean delay at the signals

4.58 → 0.60
stops per vehicle



Road damage · /roads

YOLOv8n fine-tuned on RDD2022 (4 classes), in the browser, on phone photos. A 300-image sample of the RDD2022 test split.

0.66
mAP50

66%
precision

ALL NUMBERS ARE ASSUMPTIONS TO CHECK WITH THE CUSTOMER

Who pays, and the pilot

Car park operators

Gets: free spaces now and on arrival, drivers sent to the car park with room.

Price: per camera per month, set in the pilot.

Value a month = extra spaces filled × hours × tariff, plus drivers no longer turned away at the gate.

The city

Gets: a district view from cameras it already has; less searching traffic, counted in vehicle-km.

Later: the signal audit and green wave on the same cameras.

Value: vehicle-km and minutes saved, measured in the pilot.

Outdoor advertising

Gets: vehicles past each billboard by type, direction and hour, with a log and CSV.

Price: per billboard per month, set in the pilot.

Value: counted traffic to show advertisers, from a camera or a phone video.

3 months

pilot with one operator, on the cameras it already has

People: 3 engineers part time. The main cost.

Hosting: static, free to USD 20 a month. No GPU server.

Hardware: no sensors in the ground; one camera covers the spaces it sees.

Payback = price ÷ value a month.

We state it from the pilot inputs, not before. Our cost per camera: outlining the spaces once, or checking the suggested ones, and a sample of frames by hand.

Team PariVision

Amal Karimov

Captain. Scene layout and the signal and vehicle rules

Komronbek Qodirov

Pipeline, live demo and website

Aziza Adizova

Dev set, pedestrian rules and report

Our ask

1. A 3-month pilot with one car park operator, on its own cameras.
2. One outdoor advertising company to test the billboard counter on its sites.
3. Frames from live cameras in Tashkent, to replace the recorded mornings.

Next steps

- Night test: PKLot has no night frames. A test set of our own, with hand labels.
- A hand count for the billboard counter.
- Then the same cameras time the lights: audit and green wave.

Thank you. Questions?

Try it: https://oqim-audit.vercel.app/tracker Code: https://github.com/baytoooo/oqim

Accuracy, and how each number was measured

What	Checked on	Result	How fair the check is
Spaces marked, /parking	240 held-out PKLot images, 13,447 spaces	97.4% right; precision 99.1%, recall 95.3%	Held out. Rule picked on 96 other images (97.8% there)
Same, by weather	sunny 91, cloudy 87, rainy 62 images	97.0% / 99.0% / 96.1%	Held out. No night frames in PKLot
Tracker readings	112 frames, 6,844 readings, 3 mornings	98.8% right; free count off by 1 (median), 4 (90th pct.)	Same rule as /parking, picked on its calibration images
Forecast, 10 min	101 forecasts	3.0 spaces off; "same as now" 4.6	Typical day from other weekdays; against "same as now"
Whole car park	142 labelled spaces cars used	139 found; +198 outside the labels	Checked on the labelled part only
Spaces from a photo	336 PKLot images	taken found 88.4%; suggested free right 83.5%; free found 12.2%	Settings picked on these same images. Not held out
Billboard counter	2 sample clips, midday and dusk	185 vs 189 vehicles (97.9%)	Browser YOLO26s against our offline YOLO26m. No hand count yet
Signal audit replay	4 clips, one junction	departure error 0.29 to 0.55 s on 3 clips	C3905 4.54 s: too short, fails
Road damage, /roads	300-image sample, RDD2022 test	mAP50 0.66, precision 66%	Model trained by others; its card says mAP50 0.588 on the full split

Green: held out. **Grey:** checked, but not on a held-out set. **Amber:** picked on the same data, or two models agreeing: a sanity check.

What is measured, and what is simulated

Measured, from real data

- PKLot frames and hand labels: 3 car park mornings on /tracker, 336 images on /parking.
- Kwun Tong: 22 Transport Department cameras, 1,116 snapshots counted by YOLO26m; live vacancy of 28 car parks.
- Signal audit: vehicle tracks and lamp readings on 4 clips of the organisers' footage.
- Billboard counter: vehicles crossing the line in the video, checked on 2 clips against our offline pipeline.
- Road damage: mAP50 on 300 RDD2022 test images.

Simulated or assumed

- The drivers on /city: origins and departure times are seeded.
- The trip times on /tracker (6 + 2, 8 + 4, 9 + 5 min drive + walk): example times you can change; the 40 drivers too.
- The traffic on /network: Intelligent Driver Model, one direction; the real offsets are unknown to us.
- The daily figure on /banner assumes every hour is like the one in the video.
- The predictive signal mode assumes the next 30 s of arrivals are known.
- All business numbers: assumptions to check with the customer.

Privacy, night, rain and other limits

Privacy

- No number plates and no faces are read. The detector gives vehicle boxes only.
- Photos and videos stay in the browser tab: they are not uploaded (/parking, /banner, /audit).
- A car park would publish counts only: free spaces per car park, every minute. No video is stored or sent.
- Hong Kong snapshots are public 320 x 240 images; the Transport Department greys out all but the road.

Limits

- Night: not measured. PKLot has no night frames; a hand-labelled night set is our next step.
- Rain: measured on 62 rainy held-out images, 96.1%, against 99.0% cloudy and 97.0% sunny.
- Far rows: PUCPR 96.1% (small cars at the back) against UFPR04 99.6% and UFPR05 99.3%.
- Small, masked Hong Kong snapshots: the counts are a lower bound.
- Long queues hide vehicles behind each other; the camera must not move.

Cost and speed: in the browser, no GPU server

10 MB
YOLO26n for photos and car parks, in ONNX Runtime Web

38 MB
YOLO26s for video: billboard counter and signal audit

0
GPU servers. Inference runs on the visitor's device; the site is static files.

USD 0 to 20
hosting a month, static (assumption to check)

Speed

- Video: about 1 minute of processing per minute of 1080p video on our Apple M5 laptop with WebGPU; slower on WebAssembly and on phones.
- Video is analysed at 5 frames a second.
- A photo: the whole frame plus 3 x 3 tiles, 10 detector runs; an uploaded photo adds 5 x 5 tiles over the far rows. The page prints the time it took.
- A car park needs one image a minute, not video.

Cost per camera

- No sensor in each space: one camera covers the spaces it sees.
- Setup: the spaces are suggested from one photo and checked by hand once.
- Running: the operator's own PC or a browser tab; no cloud GPU bill.
- The main cost is people: 3 engineers part time for the pilot (assumption).

How it scales to Tashkent

13,208

paid street spaces on 86 streets planned by Poytaxt Parking; paid in an app since 8 May 2025, 5,000 to 12,000 sum an hour (Spot.uz)

~1,000

signalised junctions; 125 on the adaptive control system of TsODD, 545 planned for 2026 (Anhor.uz, Yuz.uz)

309

signalised junctions and 519 km of road of central Tashkent already in the app, from OpenStreetMap

What a new car park or district needs

- One fixed camera that sees the spaces. Spaces suggested from one photo, checked by hand once.
- One image a minute, read in a browser tab or on the operator's PC. The free count goes to drivers.
- The district model is the one on /city: in Kwun Tong the vacancy comes from DATA.GOV.HK; in Tashkent it would come from the operators' cameras.
- Billboards: any fixed camera or a phone video, one line per billboard.
- Elsewhere it works: SFpark cut cruising for parking by 50% on its pilot blocks; a Copenhagen green wave cut travel time by 17% on one street (sources on /elsewhere).

Reused from the elimination, built in the final

Reused, declared

- the in-browser detector and the ONNX worker
- the ByteTrack port
- the video conversion with ffmpeg.wasm
- the site design: tokens, layout and components
- the tracks and lamp readings of the sample clips (YOLO26m at 1280 px, 10 fps)

github.com/baytoooo/parivision-traffic

Built during the final

- the parking tracker: replay, whole car park, forecast, advice for one and for many drivers
- the parking page: the space rule and its held-out test, spaces suggested from a photo
- the district model: Kwun Tong cameras, live car park vacancy, drivers spread by cost
- the billboard counter: line crossings by type and direction, log and CSV
- the signal audit core and page, the 3D corridor, the road damage page
- the camera collectors and counters, the Elsewhere page, this deck